Sahaj Bhadja

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Summary

Software Developer with 2+ years building **distributed data systems**, concurrent pipelines, and real-time applications using Python, C++, Java, SQL, cloud infrastructure, algorithms, performance optimization, and rigorous automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Operating Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Data Retrieval, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected a multi-tenant data platform using Azure SQL and Blob Storage with normalized schemas and row-level security for 500+ users, reducing analytical dashboard latency from 1.2 seconds to 350 milliseconds.
- Built FastAPI data integrations across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with incremental updates, reconciliation, failure isolation, and automated releases sustaining 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Designed a concurrent batch-processing pipeline with multithreading and secure multi-format ingestion, executing 1K+ files per batch with fault-tolerant recovery, usage tracking, and stable performance under peak load.
- Built Python services and responsive React interfaces for production workflows serving 150+ instructors and 10K+ daily requests, reducing turnaround time 60% while maintaining dependable API responsiveness during peak periods.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Optimized large-scale API integrations using object-oriented design, AWS Glue, and S3, improving response times 30%, increasing ETL pipeline efficiency 68%, and reducing technical debt 20% across production workloads.
- Engineered cross-platform data-sharing software with automated builds and 20+ unit and integration test suites, validating synchronization across three operating systems while collaborating with Microsoft and GitHub engineers.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight MERN microservices for authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across distributed workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ build validation while integrating operational monitoring and analytics across controlled staging and production environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built cross-platform booking workflows with Flask APIs, Firebase synchronization, payments, and webhooks, processing hourly Databricks micro-batches across 50K+ records with consistently reliable sub-200-millisecond responses across mobile and web clients.
- Designed end-to-end data flows, mapped 12+ integration points, and presented system architecture to stakeholders, reducing design errors 25% while supporting effective cross-functional implementation planning and delivery under evolving requirements.

Skills

Programming & Fundamentals: Python, C++, Java, JavaScript, TypeScript, SQL, Object-Oriented Programming, Data Structures, Algorithms, Operating Systems, Concurrency
Data & Distributed Systems: PostgreSQL, MySQL, SQL Server, MongoDB, Databricks, ETL, Data Modeling, REST APIs, Microservices, WebSockets, Real-Time Processing
Cloud & Infrastructure: AWS, Azure, GCP, Docker, Kubernetes, Linux, Jenkins, GitHub Actions, CI/CD, Firebase, Performance Monitoring
Engineering: System Design, Performance Optimization, Unit Testing, Integration Testing, Debugging, Reliability, Agile/Scrum, Git, Statistical Analysis, Problem Solving

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid, WebSockets*

- Built a scalable financial data platform supporting 50K+ users and 1M+ transactions with secure bank linking, real-time synchronization, advanced filtering, tokenized access, and Kubernetes-based deployment automation for reliable operational visibility.

CPU Scheduling Simulator Live | *C++, Algorithms, Operating Systems, React.js*

- Implemented six scheduling algorithms with interactive controls, dynamic waiting-time and turnaround-time calculations, and Gantt visualization, rigorously testing correctness across 1K+ preemptive and non-preemptive process combinations and workload distributions.

Linguistix | *Python, AWS, Docker, CodePipeline, ECS Fargate*

- Developed and deployed a containerized Python data-processing API on AWS using CodePipeline and ECS Fargate with automated releases, validation checks, 87% accuracy, and dependable production-grade operational reliability.